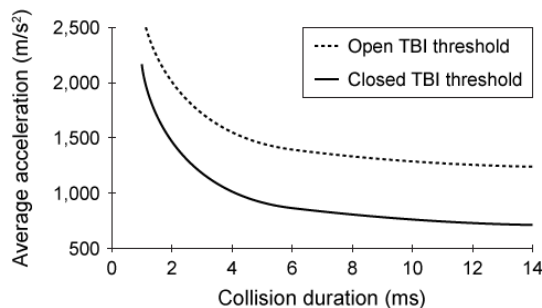


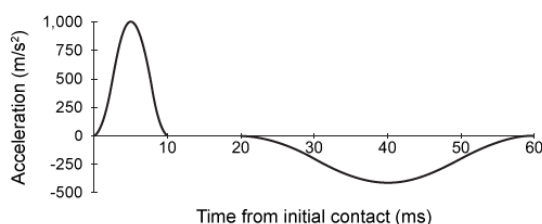
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**Figure 1** Thresholds for open and closed TBI

In an experiment to study head collisions with projectiles, researchers equipped the head of a crash test dummy with sensors that measured acceleration on multiple axes. The crash test dummy was stationary before its head was subjected to controlled collisions with incoming projectiles of varying mass and impact velocity. The linear acceleration of the crash test dummy's head was recorded during and after the impact. The results for a trial in which the impact lasted for 10 ms are shown in Figure 2.



**Figure 2** The acceleration of the head during and after a 10-ms collision

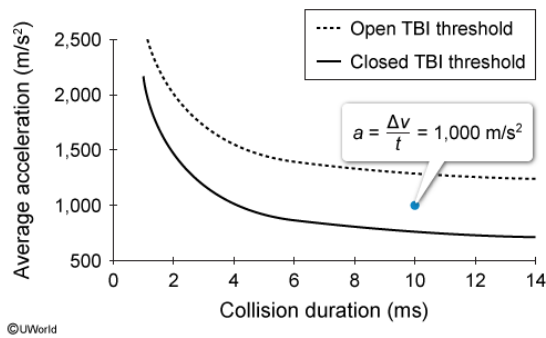
Suppose the physical characteristics of the crash test dummy accurately represent those of a human. What is the expected injury from an impact lasting 10 ms that accelerated the head to a velocity of 10 m/s?

- ☐ A. No injury [20%]
- ✓ ☒ B. TBI and no skull fracture [69%]
- ☐ C. Skull fracture and no TBI [2%]
- ☐ D. TBI and skull fracture [8%]

Correct

70% answered correctly

## Explanation:



According to the passage, the type of injury can be determined from the collision duration and the average acceleration of the head during the collision. The **average acceleration** ( $a$ ) of an object is its change in velocity ( $\Delta v$ ) over the time ( $t$ ) the object is accelerated:

$$a = \frac{\Delta v}{t}$$

In this question, the head accelerated from a stationary position to a velocity of 10 m/s ( $\Delta v = 10 \text{ m/s}$ ), and the collision lasted for 10 ms ( $t = 10 \text{ ms}$ ). The value of  $t$  is converted from **milliseconds (ms) to seconds (s)** using the conversion  $1 \text{ s} = 1,000 \text{ ms}$ :

$$t = (10 \text{ ms}) \left( \frac{1 \text{ s}}{1,000 \text{ ms}} \right) = 0.01 \text{ s}$$

Using these values, the average acceleration of the head was

$$a = \frac{\Delta v}{t} = \frac{10 \text{ m/s}}{0.01 \text{ s}} = 1,000 \text{ m/s}^2$$

In Figure 1, the point that represents an average acceleration of **1,000 m/s²** for a duration of 10 ms lies above the threshold for a closed TBI but below the threshold for an open TBI (**Choices A and C**). Therefore, a **closed TBI** is expected, which is described in the passage as a brain injury that occurs **without a skull fracture (Choice D)**.

## Educational objective:

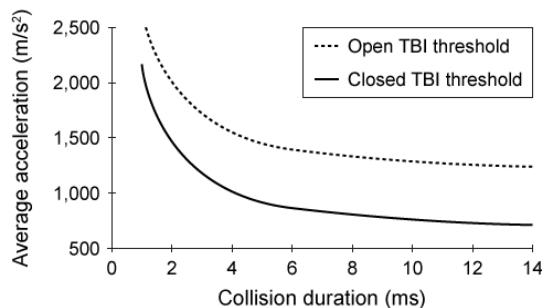
The average acceleration of an object is calculated as its change in velocity divided by the duration of the acceleration ( $a = \Delta v/t$ ).

Time spent: 0 sec  
Subject:  
Physics

QID: 400393  
System:  
Mechanics and Energy

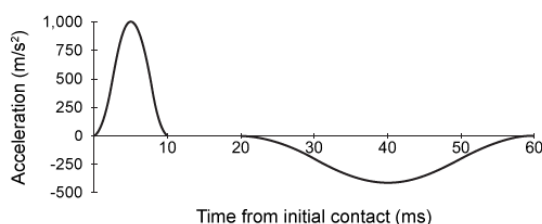
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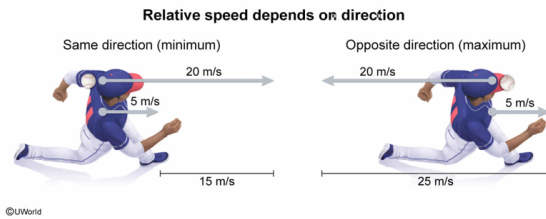


**Figure 2** The acceleration of the head during and after a 10-ms collision

A baseball traveling at 20 m/s strikes a player running at 5 m/s. The relative speed of the ball with respect to the player depends on the direction of travel of the ball relative to the player's direction of travel. What is the difference between the minimum and maximum possible speeds of the ball relative to the player?

- ✓ ☒ A. 10 m/s [74%]  
☐ B. 15 m/s [16%]  
☐ C. 20 m/s [5%]  
☐ D. 25 m/s [3%]

## Explanation:



The relative speed of the ball with respect to the player is the apparent speed of the ball from the player's perspective (also known as its frame of reference), or vice versa. The relative speed of one object with respect to another depends on their direction of motion relative to one another.

The **minimum** relative speed  $v$  occurs when the ball and player are moving in the **same direction**. In this case,  $v$  equals the difference between the speed of the ball and the speed of the runner:

$$v = 20 \text{ m/s} - 5 \text{ m/s} = 15 \text{ m/s}$$

The **maximum**  $v$  occurs when the ball and the player are moving in **opposite directions**. In this case,  $v$  is the sum of the ball and runner speeds:

$$v = 20 \text{ m/s} + 5 \text{ m/s} = 25 \text{ m/s}$$

Therefore, the difference between the minimum (15 m/s) and maximum (25 m/s) possible values is **10 m/s**.

**(Choices B and D)** The minimum and maximum possible values are 15 m/s and 25 m/s, respectively.

## Educational objective:

The relative speed between two objects depends on their directions of motion. The minimum relative speed occurs when two objects move in the same direction, and the maximum relative speed occurs when two objects move in opposite directions.

**Time spent:** 0 sec

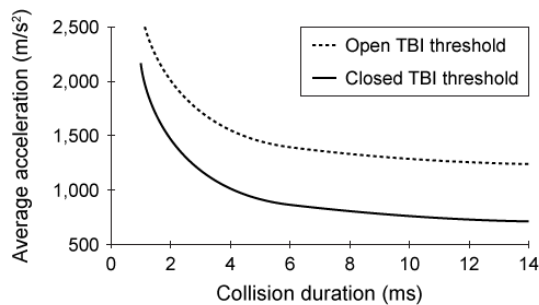
**Subject:**  
Physics

**QID:** 400394

**System:**  
Mechanics and Energy

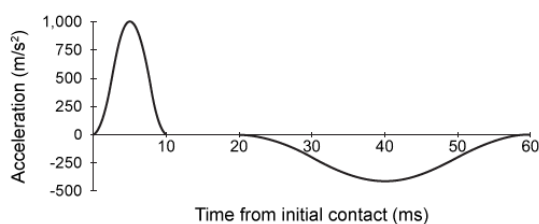
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**Figure 1** Thresholds for open and closed TBI

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**Figure 2** The acceleration of the head during and after a 10-ms collision

In the range from 0 to 10 ms in Figure 2, which of the following represents the velocity of the head at 10 ms?

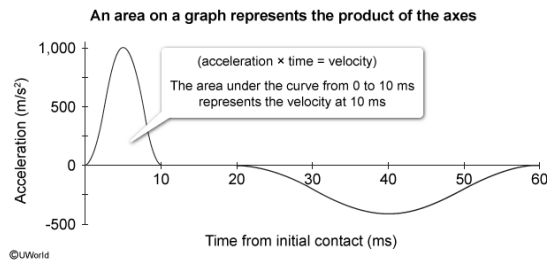
- ☐ A. The length of the curve [2%]
- ☒ B. The area under the curve [81%]
- ☐ C. The maximum height of the curve [3%]
- ☐ D. The maximum slope of the curve [13%]

Correct

80% answered correctly

Explanation:

## Explanation.



In Figure 2, the acceleration of the dummy's head ( $y$ -axis) is given as a function of time ( $x$ -axis). **Acceleration** ( $a$ ) is the rate at which velocity changes, and it is calculated as the change in velocity ( $\Delta v$ ) divided by the duration ( $t$ ) of the acceleration:

$$a = \frac{\Delta v}{t}$$

Therefore, the **change in velocity** of the head can be determined as the product of acceleration and its duration:

$$\Delta v = at$$

On a graph, the **product of the axes' quantities** is represented by the area under a curve. Therefore, the area under the curve between two points in time in Figure 2 represents the  $\Delta v$  of the head during that time interval. Because the velocity of the head is 0 m/s at the start of the graph, the **area under the curve** from 0 to 10 ms represents the velocity of the head at 10 ms.

**(Choice A)** The length of the curve does not represent an important physical quantity.

**(Choice C)** The maximum height of the curve represents the maximum acceleration of the head.

**(Choice D)** The slope (change in the  $y$ -axis over the change in the  $x$ -axis) of the acceleration vs. time graph in Figure 2 gives the rate at which acceleration changes per unit time, not velocity. The slope of a distance vs. time graph would represent velocity ( $v = d/t$ ).

### Educational objective:

Acceleration is the rate at which velocity changes over time, and therefore the product of acceleration and time is the change in velocity. On a graph, the product of the axes' quantities is represented by the area under a curve. Therefore, the area under the curve on an acceleration vs. time graph represents the change in velocity.

Time spent: 0 sec

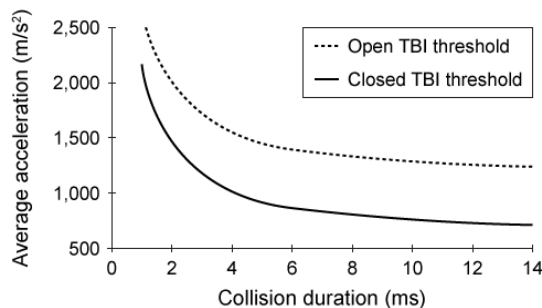
Subject:  
Physics

QID: 400395

System:  
Mechanics and Energy

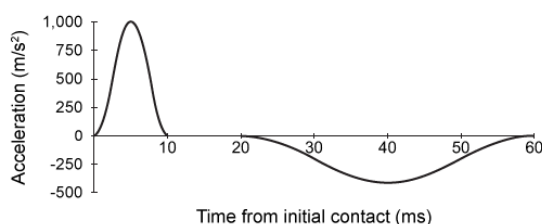
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**Figure 2** The acceleration of the head during and after a 10-ms collision

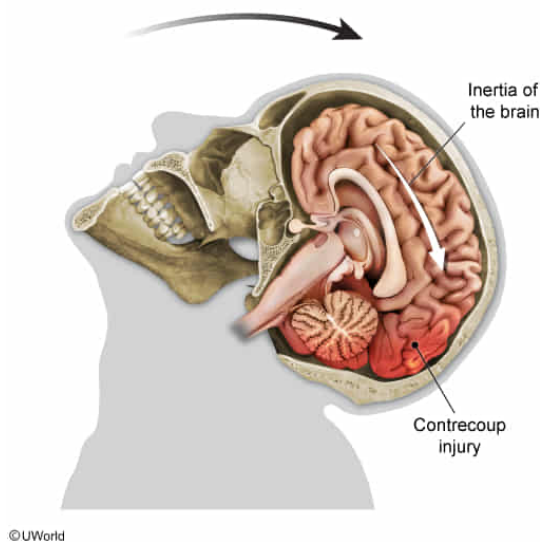
When a stationary head is hit by a moving projectile, a contrecoup injury is likely to occur due to:

- ✓ ☒ A. the inertia of the brain. [80%]
- ☐ B. the weight of the skull. [2%]
- ☐ C. the center of mass of the head. [8%]
- ☐ D. the potential energy of the projectile. [8%]

Correct

80% answered correctly

## Explanation:



A contrecoup injury is described in the passage as a brain injury that occurs on the side of the head opposite an impact. This form of brain injury involves the **movement of the brain relative to the skull** due to the rapid acceleration and deceleration of the head.

When a stationary head is hit by a moving object, the skull and the brain are both accelerated in the same direction. Because the brain can move within the skull, the **brain will continue to move** due to its inertia even after the skull had slowed significantly or stopped. An object's **inertia** is proportional to its mass, and it is the tendency of an object to resist changes to its speed; an object in motion will stay in motion, and an object at rest will stay at rest ([Newton's first law](#)). Therefore, the inertia of the brain can lead to a brain-skull collision on the opposite side of the head from the initial impact (contrecoup injury).

**(Choice B)** [An object's weight](#) is the gravitational force exerted on its mass. The weight of the skull does not cause the continued relative motion of the brain within the skull.

**(Choice C)** An object's center of mass is the average position of its distribution of mass. The head's center of mass changes as a result of the movement of the brain within the skull, but the center of mass itself is not responsible for the movement of the brain.

**(Choice D)** An object's potential energy is its "stored" energy due to its height above the ground. The potential energy of the projectile does not cause the continued movement of the brain relative to the skull.

## Educational objective:

An object's inertia is its resistance to changes in its velocity; objects tend to stay in motion or stay at rest. Due to the brain's inertia, the brain can continue to move independently of the skull and result in a contrecoup injury.

Time spent: 0 sec

Subject:  
Physics

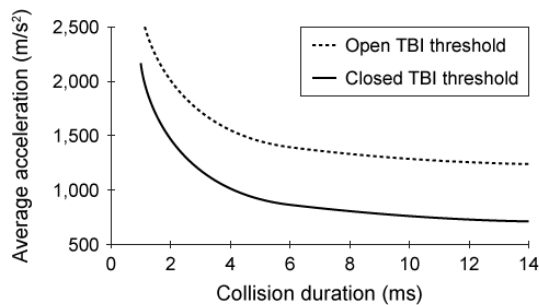
QID: 400396

System:  
Mechanics and Energy



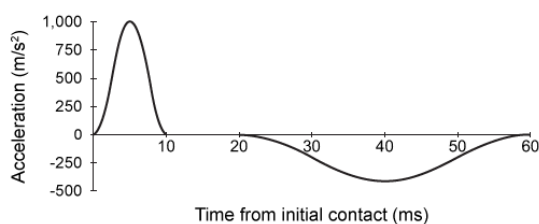
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**Figure 2** The acceleration of the head during and after a 10-ms collision

What additional information do the researchers need from the crash test dummy to estimate the power of the collision in the experiment shown in Figure 2?

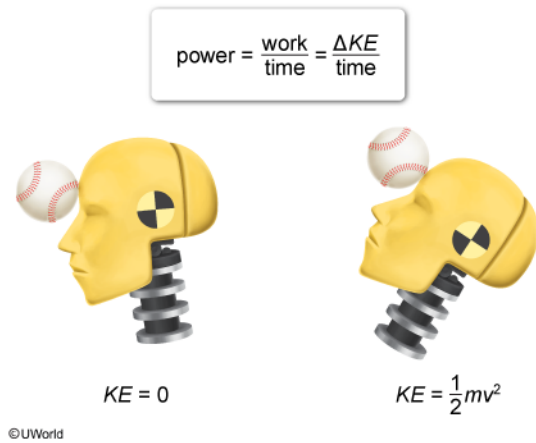
- ✓ ☒ A. Mass of the head [72%]
- ☐ B. Time until the head stops moving [11%]
- ☐ C. Tension of the neck during the collision [7%]
- ☐ D. Velocity of the head at the end of the collision [8%]

Correct

72% answered correctly

Explanation:

### Explanation:



**Power** ( $P$ ) is the rate of work ( $W$ ) per unit time ( $t$ ), and work is the energy transferred by a force through a distance. From the work-energy theorem, **work** is equal to the **change in the kinetic energy** ( $KE$ ) of an object. Therefore, power can be calculated from  $\Delta KE$  and  $t$ :

$$P = \frac{W}{t} = \frac{\Delta KE}{t}$$

An object's **kinetic energy** is the energy associated with its motion, and it is calculated from its mass  $m$  and velocity  $v$ :

$$KE = \frac{1}{2}mv^2$$

Therefore, the power of the collision shown in Figure 2 can be determined from its  $\Delta KE$ , which depends on the head's mass and its initial and final velocities during the collision. The head is stated to be stationary initially, and the final velocity of the head can be determined from the information given in Figure 2 (**Choice D**). Therefore, the **mass of the head** is needed in addition to the given information to determine the power of the collision.

**(Choice B)** The power of the collision is determined from the *duration of the collision*. The time until the head stops moving may occur after the collision has ended.

**(Choice C)** Tension is the force experienced by a string (or similar object) due to opposing forces at each of its ends. The forces at the neck do not need to be known to estimate the power of the collision.

### Educational objective:

Power is the rate of work per unit time, and work is equal to the change in the kinetic energy of an object. Kinetic energy is due to motion, and it is calculated from an object's mass and velocity:  $KE = \frac{1}{2}mv^2$ .

Time spent: 0 sec

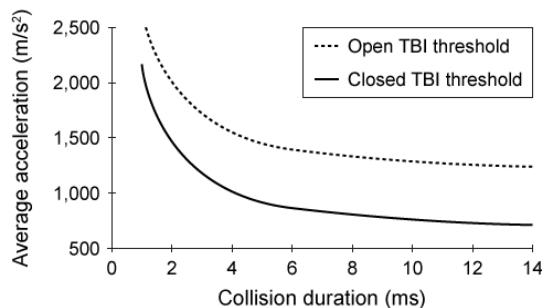
Subject:  
Physics

QID: 400397

System:  
Mechanics and Energy

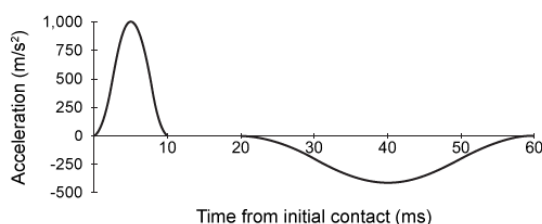
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**Figure 2** The acceleration of the head during and after a 10-ms collision

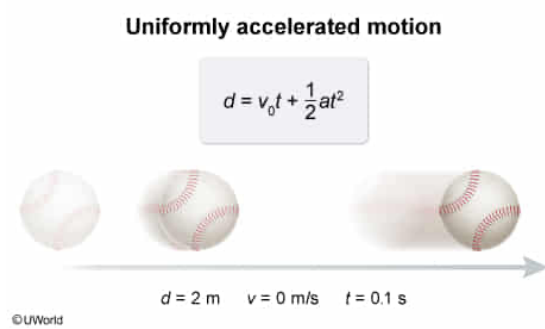
In one experimental trial, the projectile was uniformly accelerated from rest to a distance of 2 m in 0.1 s. What was the acceleration of the projectile?

- ☐ A. 20 m/s<sup>2</sup> [17%]
- ☐ B. 40 m/s<sup>2</sup> [5%]
- ☐ C. 200 m/s<sup>2</sup> [44%]
- ☒ D. 400 m/s<sup>2</sup> [33%]

Correct

33% answered correctly

## Explanation:



The distance ( $d$ ) traveled by a uniformly accelerating object is given by the **kinematic equation**:

$$d = v_0 t + \frac{1}{2} a t^2$$

where  $v_0$  is the initial velocity,  $a$  is the constant acceleration, and  $t$  is the duration of time.

According to the question, the projectile **started from rest** with an initial velocity  $v_0$  of 0 m/s.

Therefore, the equation can be simplified to:

$$d = (0)t + \frac{1}{2} a t^2$$

$$d = \frac{1}{2} a t^2$$

The **acceleration of the projectile** is determined by rearranging the equation to solve for  $a$  and plugging in the given values ( $d = 2 \text{ m}$ ,  $t = 0.1 \text{ s}$ ):

$$a = 2d/t^2$$

$$a = 2(2 \text{ m})/(0.1 \text{ s})^2$$

$$a = 400 \text{ m/s}^2$$

**(Choice A)** Velocity is calculated as distance divided by time:  $v = d/t$ . The average velocity of the projectile is 20 m/s, but the question asks for acceleration.

**(Choices B and C)** The distance traveled starting from rest is  $d = \frac{1}{2} a t^2$ . If  $t$  is used instead of  $t^2$ , the incorrectly calculated acceleration is 40 m/s<sup>2</sup>. If the  $\frac{1}{2}$  coefficient is omitted, the incorrectly calculated acceleration is 200 m/s<sup>2</sup>.

## Educational objective:

The kinematic equation that describes the distance traveled by a uniformly accelerated object is  $d = v_0 t + \frac{1}{2} a t^2$ , where  $d$  is distance,  $v_0$  is the initial velocity,  $t$  is time, and  $a$  is acceleration.

This equation can be rearranged to solve for an object's acceleration.

Time spent: 0 sec

Subject:

Physics

QID: 400398

System:

Mechanics and Energy